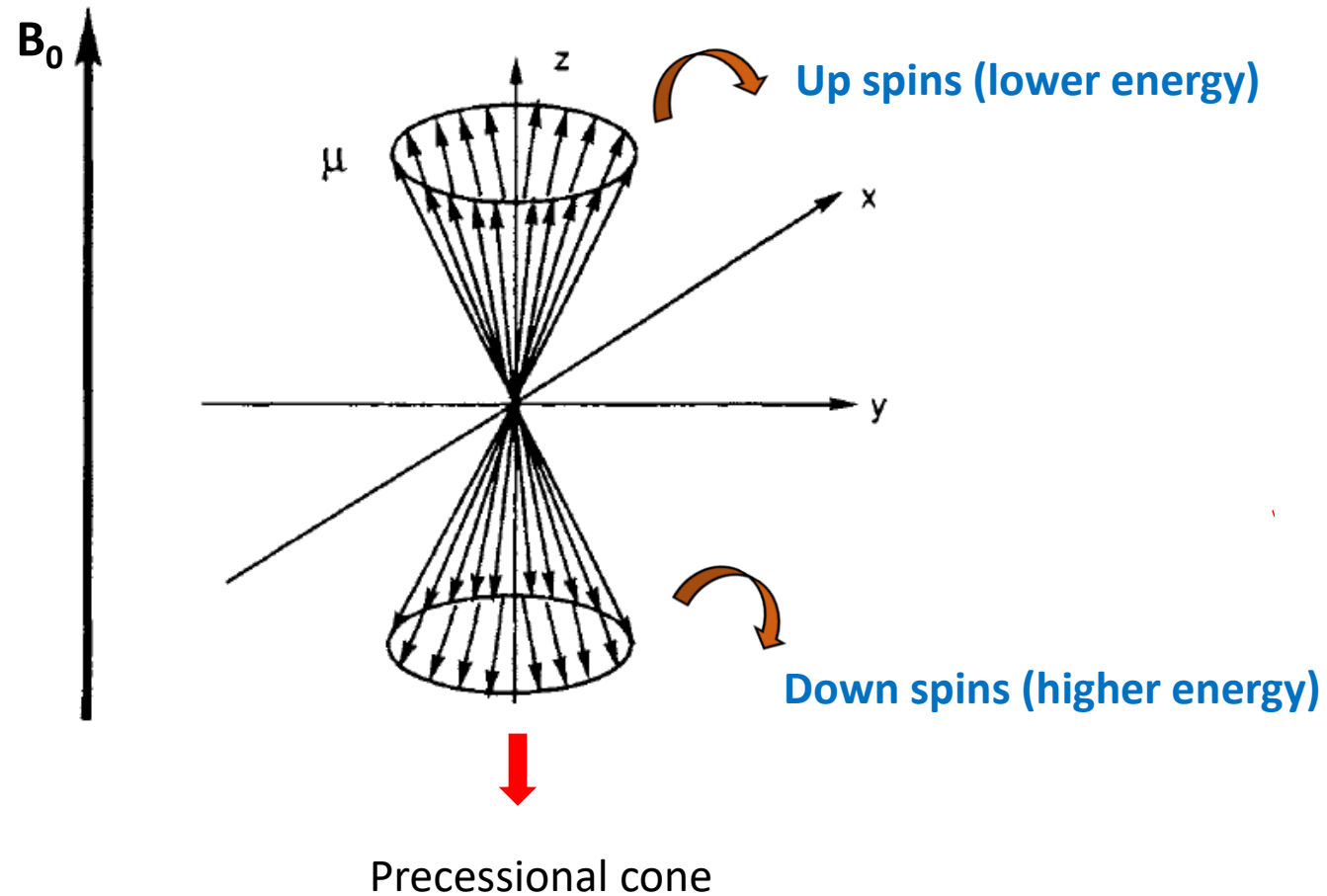
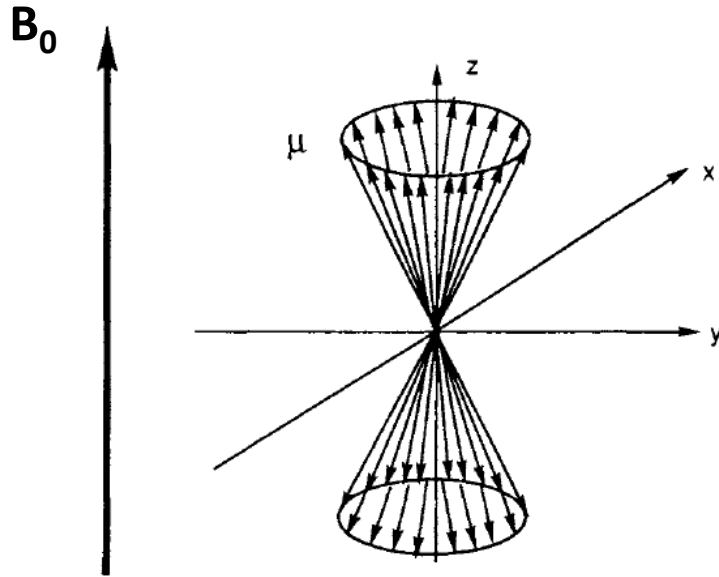


In NMR, we will always consider that the magnetic field to be applied along Z direction

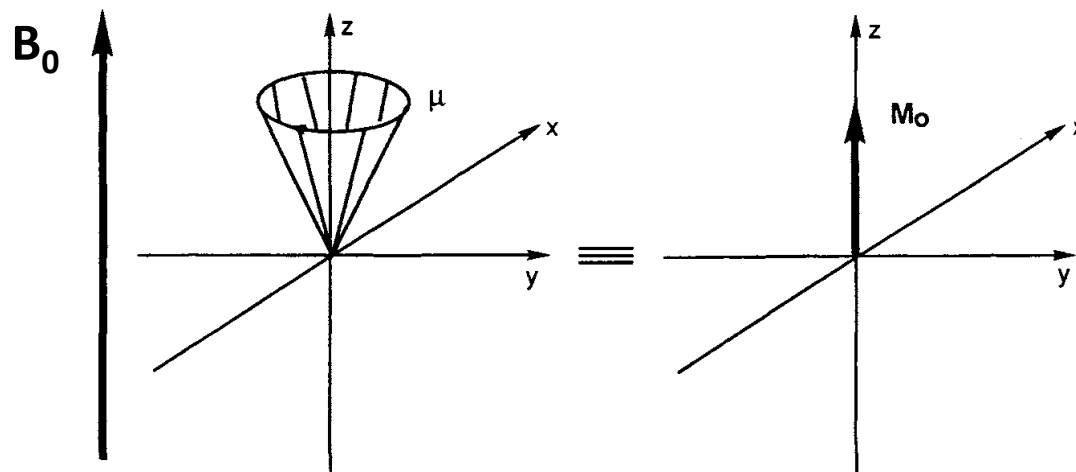


- Net magnetization (M_0) in the z direction as there are slightly more number of **up** spins than **down** spins

Magnetization Along z Direction



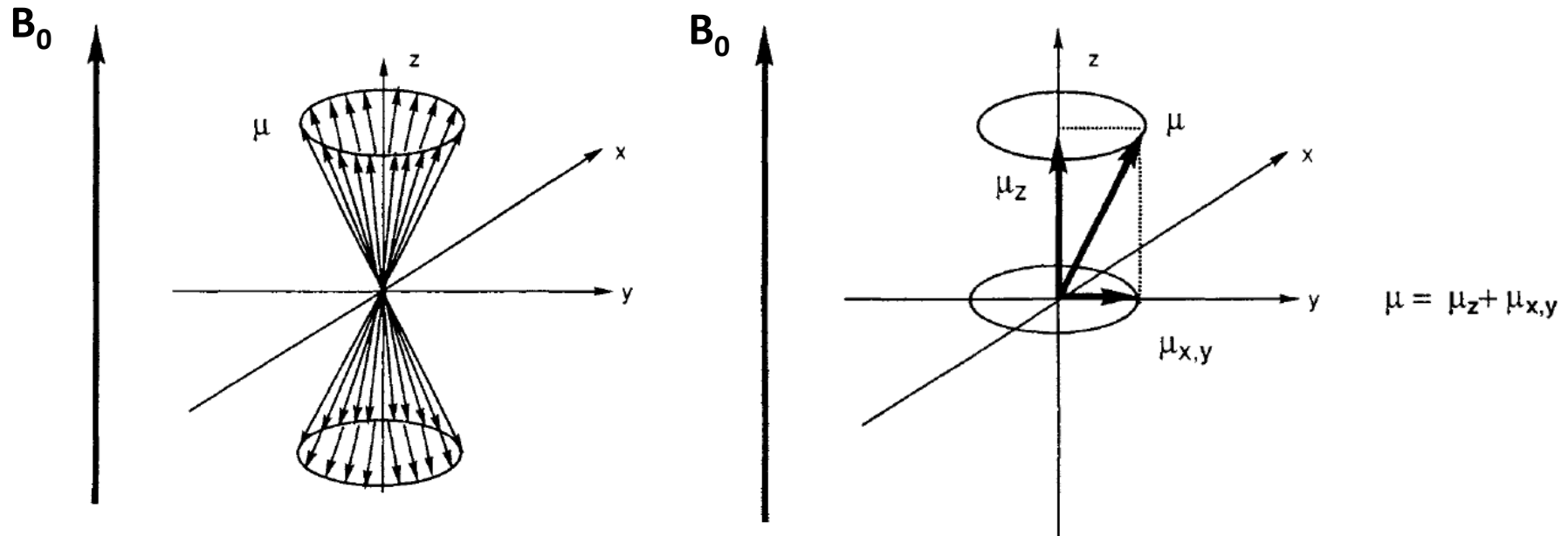
- The spins are distributed homogeneously around the precessional cone
- There are a slightly more number of nuclear spins along the magnetic field
- A net magnetization along the z axis



M_0 : Longitudinal/Bulk
magnetization

(Thermal Equilibrium)

Magnetization in XY Plane

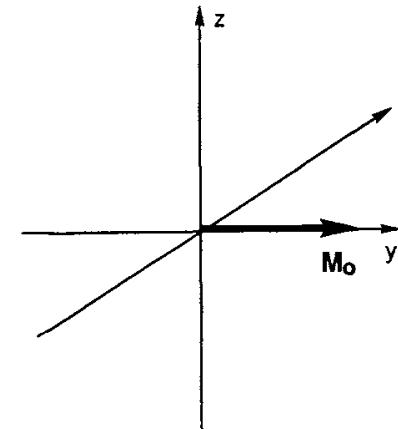
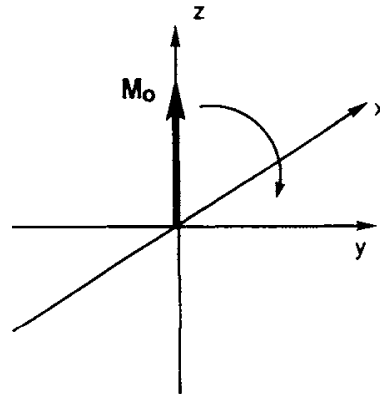


No net magnetization in the xy plane $\sum \mu_{x,y} = 0$

NMR spectroscopy takes a different approach.....

➤ Instead of measuring the magnetization along the field, the magnetization along the perpendicular direction is measured

Pulse NMR Spectroscopy

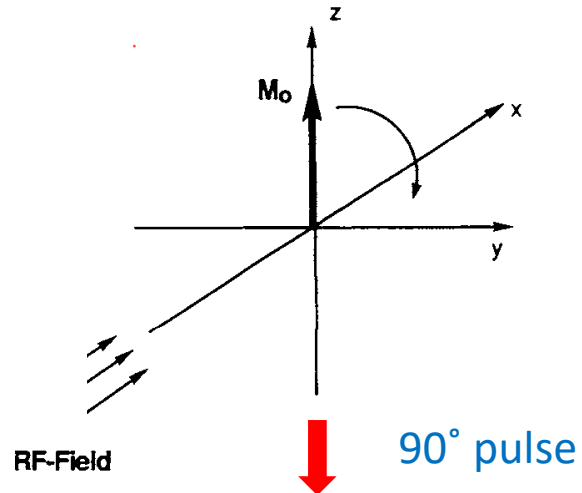


How does it happen?

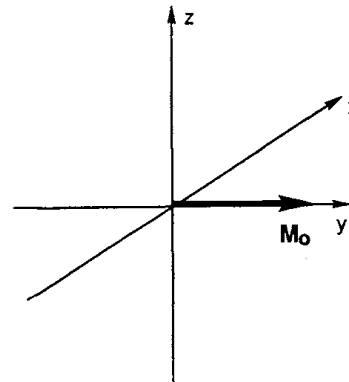
- A pulse is a burst of radiofrequency energy that may be applied by switching a R_f transmitter
- As long as the pulse is applied, a constant force (*torque*) is exerted on the sample magnetization, causing it to precess about the R_f vector

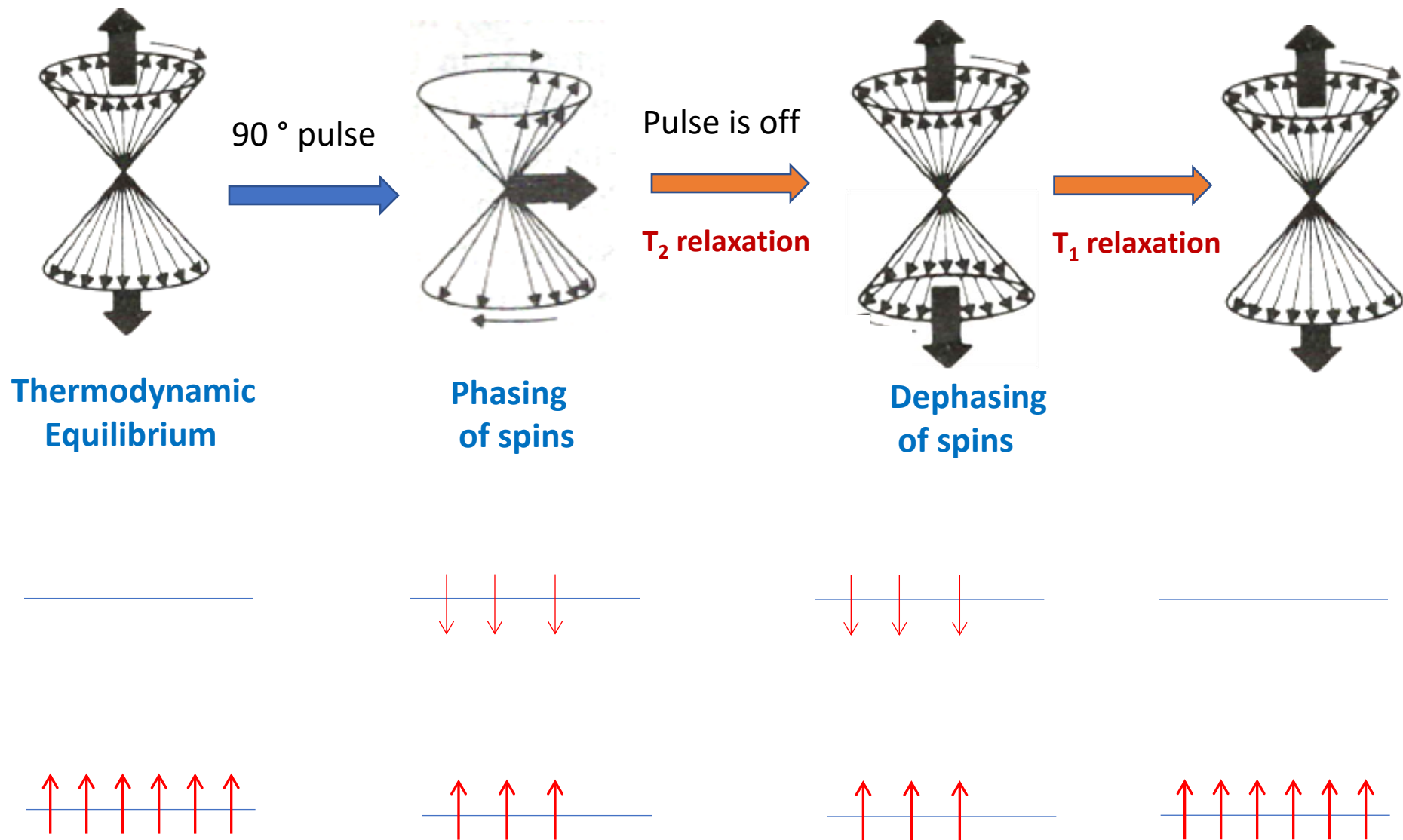


Richard Ernst
ETH Zurich
(Nobel prize 1991)



?

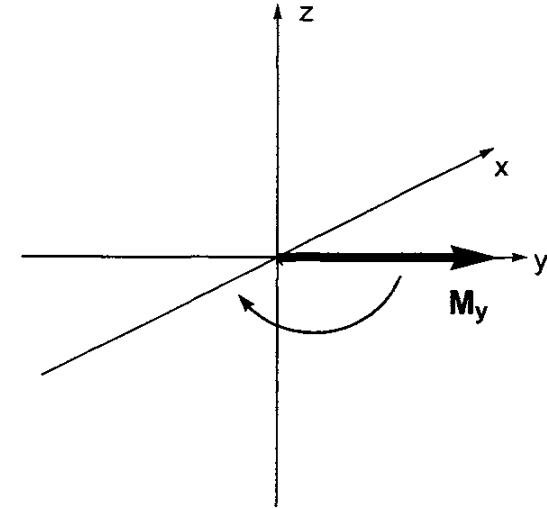




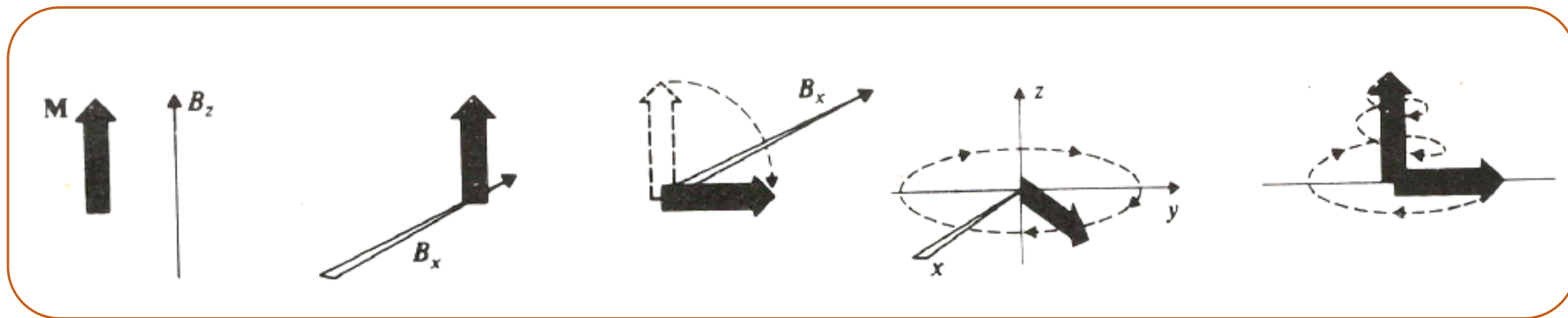
✓ In reality, T_1 and T_2 relaxations happen simultaneously

What happens immediately after the 90° pulse is switched off?

- $M_z = 0, M_y = M_0$
- Relaxation processes occur to release the excess energy

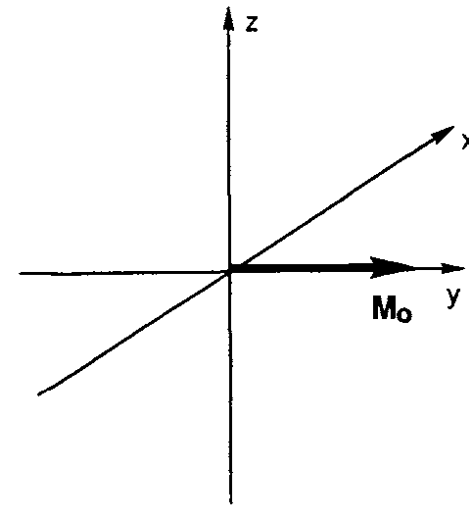
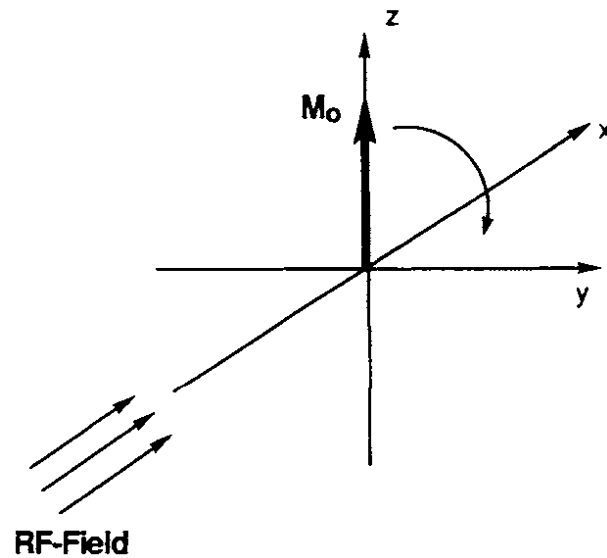


- M_y decays with time and finally $M_z = M_0$
- This relaxation process is characterized by T_2 , the spin-spin relaxation constant



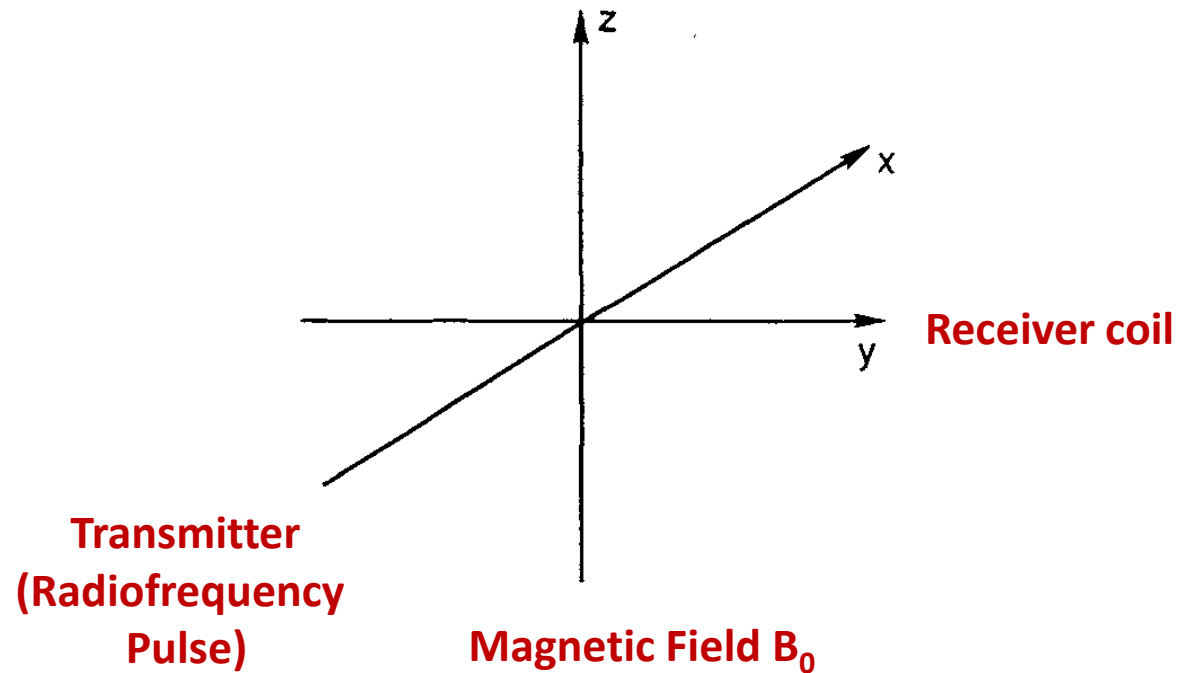
Pulse NMR Spectroscopy

- We can control the extent by which the +z magnetization is bent by choosing the duration for which the pulse is applied
- The term **90° pulse** actually means the time period for which the pulse has to be applied



The modern day NMR spectrometers basically do 3 things

1. Magnetizing the nuclear spins with a large magnetic field
2. Rotating the spin polarizations to produce transverse nuclear magnetization
3. Detecting the small oscillating electric currents induced by precessing transverse spin magnetization



Attached Proton Test (APT)

➤ Distinguishes C and CH₂ from CH and CH₃

Pulse sequence: J modulation-Spin Echo

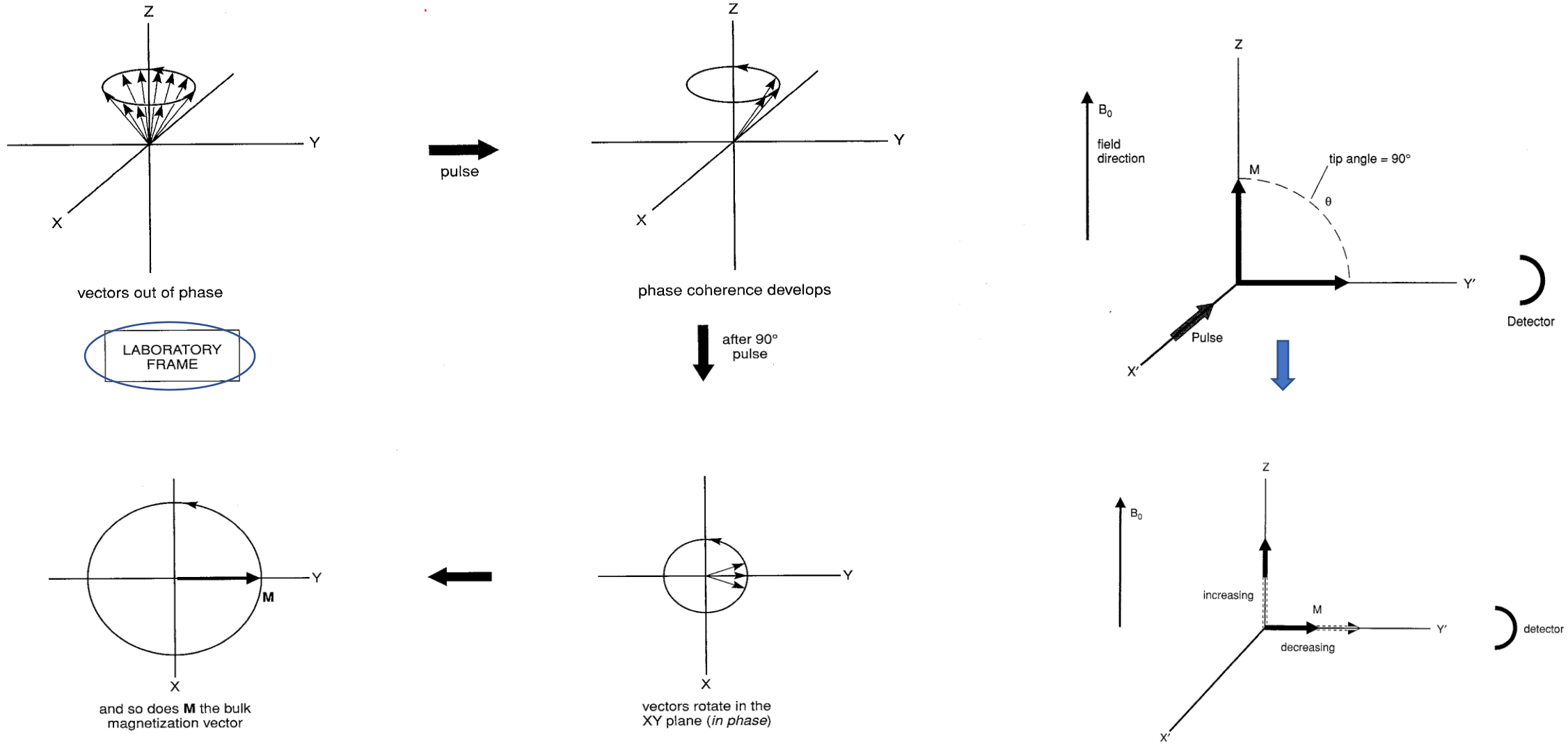
¹³C channel: 90°_{X'} - 1/J - 180°_{X'} - 1/J - Observe

¹H channel: BB decoupling

➤ CH₃, CH: positive peak

➤ C, CH₂: negative peak

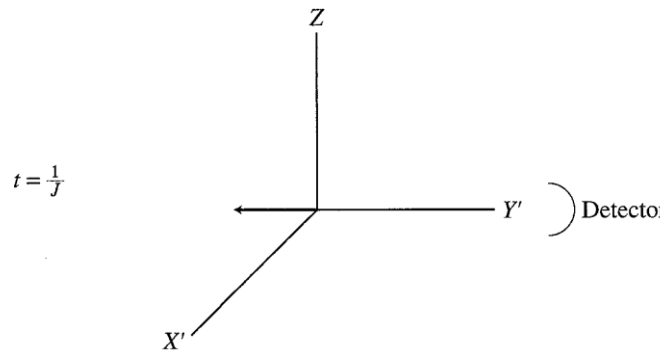
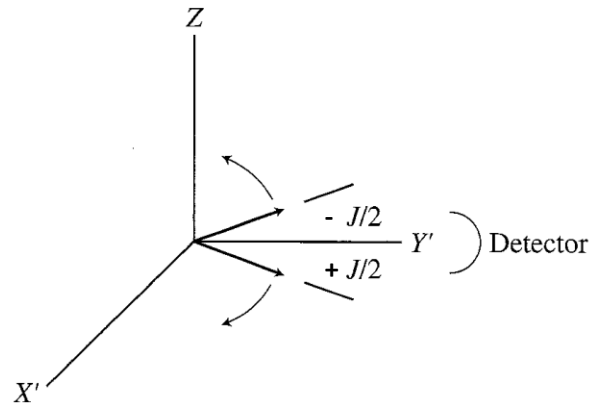
Rotating Frame of Reference



➤ A set of coordinate axes (X' , Y') that rotate in the same direction and at the same Larmor frequency of the spins 11

The DEPT experiment is a modification of a basic NMR experiment called the **attached proton test (APT)** experiment. Although a detailed explanation of the theory underlying the DEPT experiment is beyond the scope of this book (Pavia), an examination of APT is given below

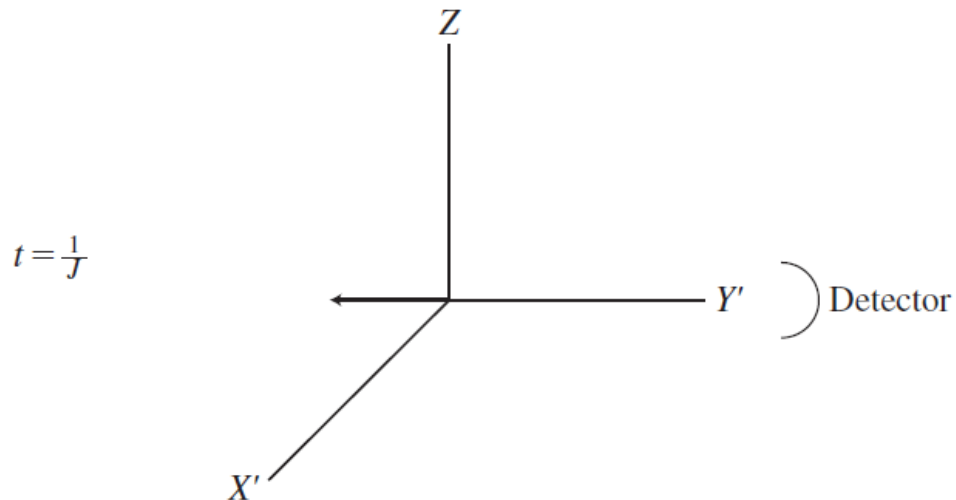
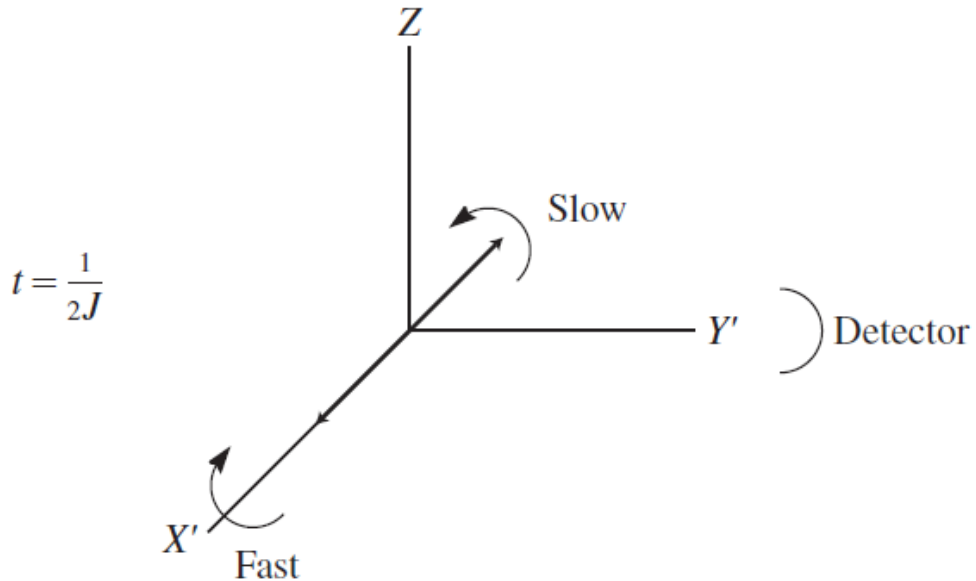
Methine Carbons (CH)



In this case, the attached hydrogen splits this resonance into a doublet. The resonance does not occur exactly at the Larmor frequency; rather, coupling to the proton produces *two* vectors. One of the vectors rotates $J/2$ Hz *faster* than the Larmor frequency, and the other vector rotates $J/2$ Hz *slower* than the Larmor frequency. One vector results from a coupling to the proton with its magnetic moment aligned with the magnetic field, and the other vector results from a coupling to the proton with its magnetic moment aligned against the magnetic field. The two vectors are separated in the rotating frame.

Methine Carbons (CH)

The vectors are moving relative to the rotating frame at a speed of $J/2$ revolutions per second but in opposite directions. The time required for one revolution is therefore the inverse of this speed or $2/J$ seconds per revolution. At time $1/4(2/J)$, $= 1/2 J$ the vectors have made one-fourth of a revolution and are opposite each other along the X axis. At this point, no signal is detected by the receiver because there is no component of magnetization along the Y axis (the resultant of these two vectors is zero).

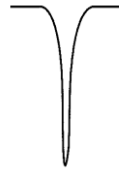
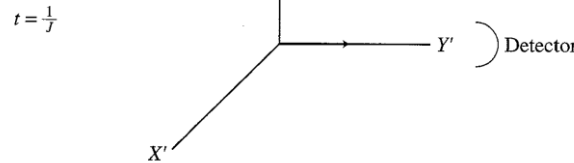
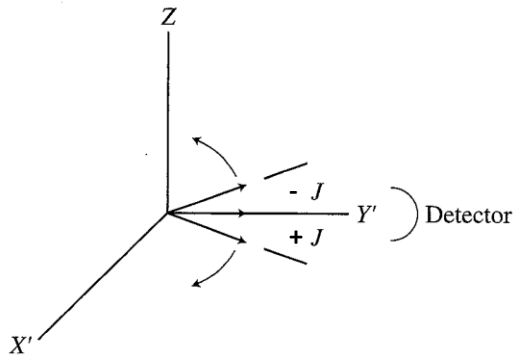
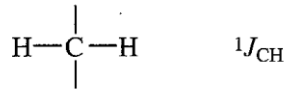


At time $1/2(2/J) = 1/J$, the vectors have realigned along the Y axis but in the negative direction. An inverted peak would be produced if we collected signal at that time. So, if $t = 1/J$, a methine carbon should show an inverted peak.

Methylene Carbons (CH_2)

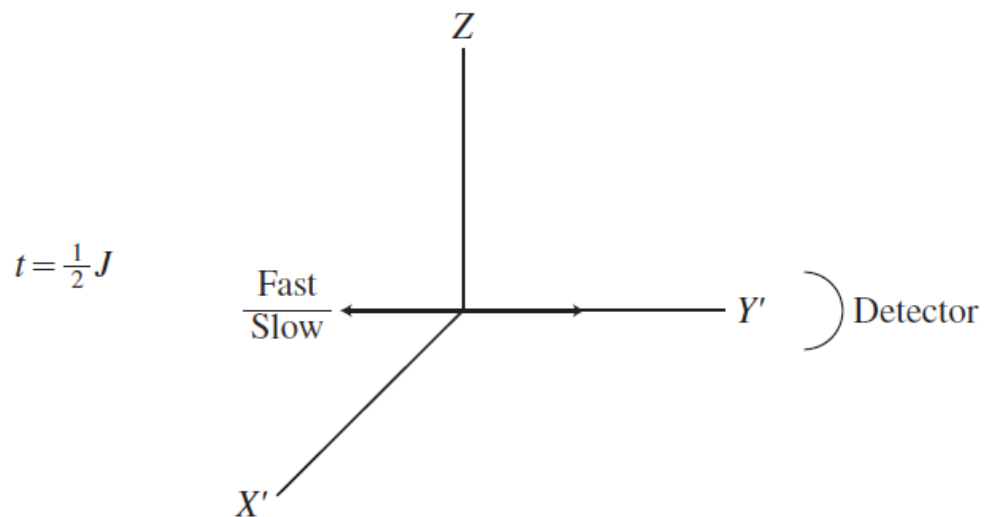
If we examine the fate of a ^{13}C atom with *two* attached protons, we find different behavior:

In this case, there are *three* vectors for the ^{13}C nucleus because the two attached protons split the ^{13}C resonance into a triplet. One of these vectors remains stationary in the rotating frame, while the other two move apart with a speed of J revolutions per second.

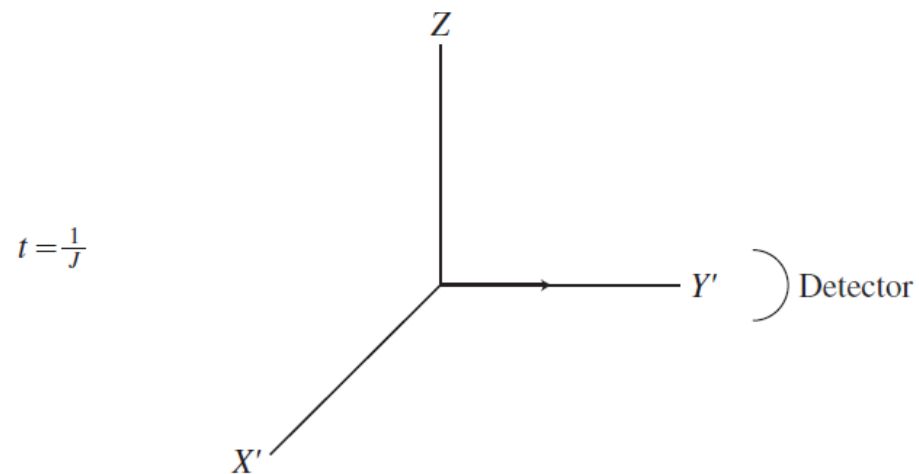


Methylene Carbons (CH_2)

At time $\frac{1}{2}(1/J)$, the two moving vectors have realigned along the negative Y' axis,

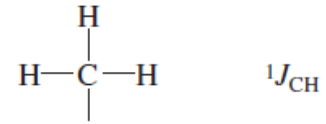
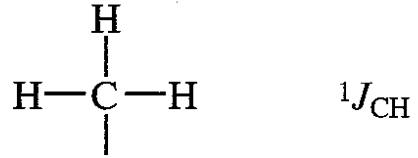


and at time $1/J$, they have realigned along the positive Y axis. The vectors thus produce a normal peak if they are detected at time $t = 1/J$. Therefore, a methylene carbon should show a normal (positive) peak.

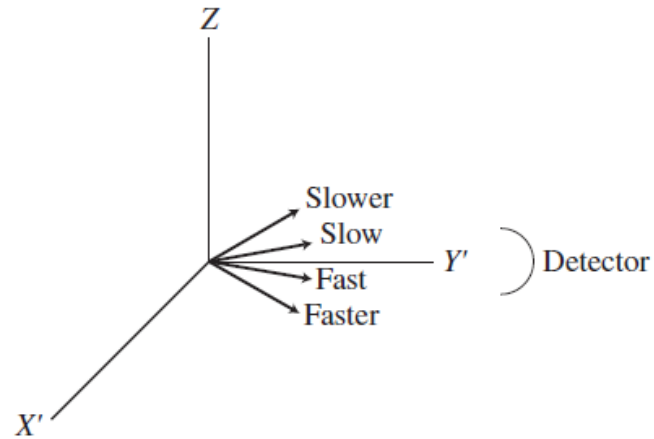


Methyl Carbons (CH_3)

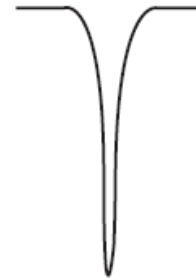
In the case of a methyl carbon,



there should be *four* vectors, corresponding to the four possible spin states of a collection of three hydrogen nuclei.



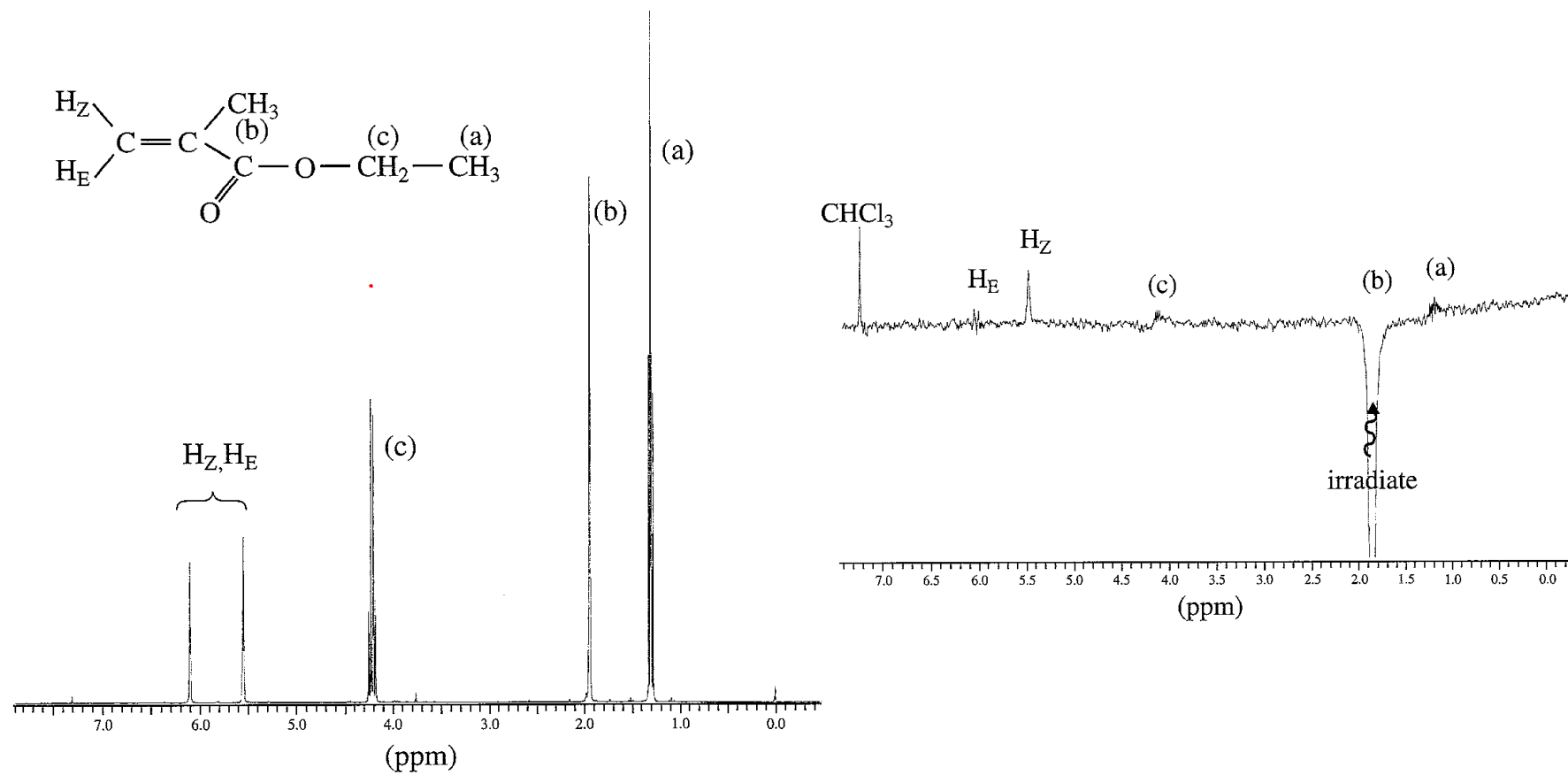
An analysis of the precessional frequencies of these vectors shows that after time $t = 1/J$, the methyl carbon should also show an inverted peak.

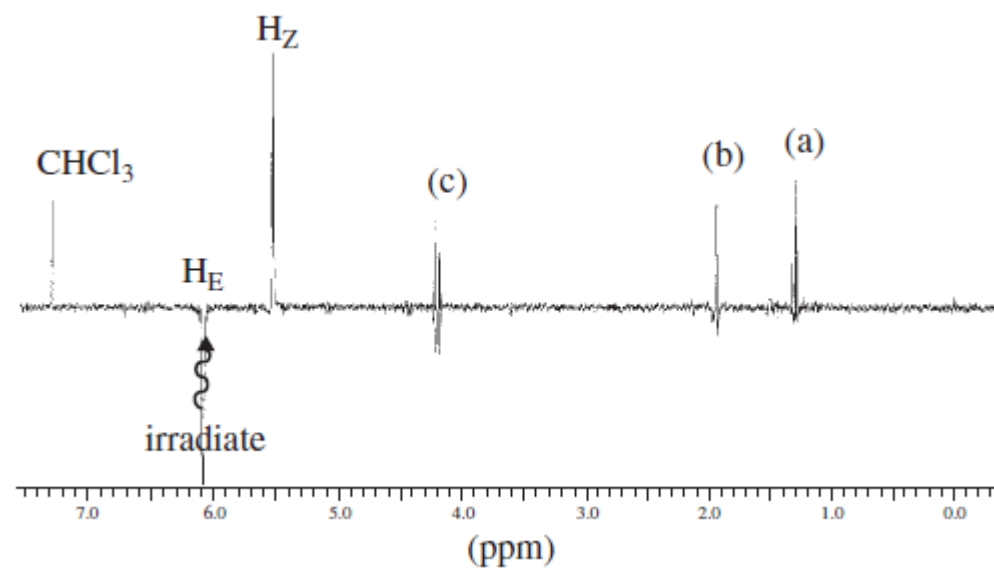


In the form of this experiment known as the DEPT experiment, the pulse sequence is more complex than those described in the preceding paragraphs. By varying the pulse widths and delay times, it is possible to obtain separate spectra for methyl, methylene, and methine carbons. In the normal manner of presenting DEPT spectra (e.g., a DEPT-135 spectrum), the trace that combines the spectra of all these types of ^{13}C atoms is inverted from the presentation described for the attached proton test (APT).

In the DEPT experiment, results similar to those described here for the APT experiment are obtained. A variety of pulse angles and delay times are incorporated into the pulse sequence. The result of the DEPT experiment is that methyl, methylene, methine, and quaternary carbons can be distinguished from one another.

NOE Difference Spectra





Problem 1: You have a benzene ring with three substituents: a Methoxy group (-OCH₃), a Nitro group and a single Bromine. You know it is a trisubstituted benzene, but you don't know the substitution pattern 1,2,3 or 1,2,4

Signal 1 (delta 3.9): Singlet, 3H

Signal 2 (delta 7.2): Doublet, 1H_a

Signal 3 (delta 6.8): Doublet of doublets, 1H_b

Signal 4 (delta 7.5): Doublet, 1H_c

You irradiate the singlet (delta 3.9).The Result:

Signal 2 shows a significant NOE enhancement.

Signal 3, 4 shows no enhancement.

Predict the structure.